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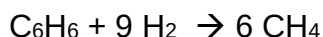
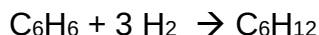
1.0 - Executive Summary

This project has the objective of simulating a Cyclohexane Production through Aspen, where design specifications were created in order to meet the target ratios and a production of 300 KMTA final product by varying the Hydrogen to Benzene ratio and the recycle stream. This part also utilizes the problem of the previous homework, however the flow diagram had to be modified to a distillation column instead of a low-pressure separator. In addition to those changes, the objective of this problem is to calculate the CAPEX (capital cost), OPEX (operation cost), including utilities and raw materials, revenues from the product and credits for fuel. In total, this plant will cost. The total cost of equipment was estimated as \$1,875.98, while the annual operating cost was of \$2,444,642 in addition to the cost of utilities, raw material and credits for product and fuel production. More details about cost breaking down and credits towards the final product and fuel produced can be found in Appendix III.

2.0 – System and Procedure

The system includes two pumps, a compressor, a mixer, a bubble column reactor, a throttling valve, two splitters, a cooler, two heaters, one separation column and one distillation column. More details can be found in the Process Flow Diagram in Appendix II.

In the original problem, benzene enters as a fresh feed at 100°F and 15 Psia by a supply stream (100 lbmol/h), followed by a secondary stream of fresh feed (313 lbmol/h) containing 97.5 % hydrogen gas (305.175 lbmol/h), 0.5% nitrogen gas (1.565 lbmol/h) and 2% methane gas (6.26 lbmol/h) at 120°F and 335 Psia. The two streams are combined with two recycling streams from the first high pressure separator, and all streams are mixed inside a mixer. The mix is then fed to the reactor where two reactions take place at 400°F and 315 Psia:



The resultant stream from the reactor is then cooled to 120°F and fed to the high-pressure separator at 315 Psia, where a vapor stream comes out of the top and a liquid stream comes out of the bottom. The top stream is followed by a purge and a recycle stream back to the mixer. The bottom stream is also split, where the resulting streams will be a liquid recycle back to the mixer and a secondary stream as the feed for the low-pressure separator. At the low-pressure separator, the final product will be collected out of the bottom stream, which may also include some impurities. The top vapor stream is composed mainly of methane.

The system presents some parameters that must be met in order to be considered a successful result. The vapor recycled stream must be adjusted in order to maintain a hydrogen to Benzene molar ratio of 3.3 in the mixer, while the liquid recycled stream must be no more than 45% of the fresh feed liquid Benzene flow rate. In addition to the original problem, the flowsheet must be reconfigured to show all the unit operations corresponding to the actual equipment in the field, as well as increasing the production of cyclohexane to 300 KMTA, while meeting all the requirements described above. A list with all the utilities and total operation cost for this product is also required, which includes the cost of raw material, credits for fuel production and revenue from the product.

For this assignment, the procedure was as follows:

1. The process flow diagram was obtained from a previous assignment.
2. Flowsheet was modified to include all the unit operations needed.
3. For the first design specification the purge stream should change in order to keep a molar ratio of 3:3 hydrogen to benzene, based on the tear stream. Since the tear stream composition is the same as the composition inside the mixer.

4. The second design specification was created around the liquid recycle, where the recycle stream was set as 45% of the fresh liquid benzene flow rate.
5. A calculator was set to calculate the fresh feed going inside the reactor based on the amount of product desired.
6. Two heaters were set as a heater exchanger.
7. A water cooler was set up in the system.
8. The low-pressure separator was switched for a distillation column.
9. Temperature of the cooling water into the cooler was calculated.
10. New set of units was created to output the flow rate as KMTA.
11. All the costs were calculated.

3.0 – Physical Properties

Property Name	Units	H2	N2	CH4	C6H6	C6H12_1	H2O
Molecular Weight		2.0160	28.0130	16.0430	78.1140	84.1620	18.0150
Critical Temperature	C	-239.9500	-146.9500	-82.5500	288.9500	280.2500	374.1500
Critical Pressure	atm	12.8000	33.5000	45.4000	48.3000	40.2000	217.6000
Critical Volume	cm ³ /mol	64.9115	89.6408	99.2094	258.7791	308.3679	55.8953
Compressibility Factor		0.3050	0.2900	0.2880	0.2710	0.2730	0.2290
Accentric Factor		-0.2200	0.0400	0.0080	0.2120	0.2130	0.3440
Normal Boiling Point	C	-252.7500	-195.7500	-161.4500	80.1500	80.7500	100.0000
Freezing Point	C	-259.1500	-209.8500	-182.4500	5.5500	6.5500	0.0000
Liquid Molar Volume	cm ³ /mol	28.6127	34.7065	37.7482	95.5084	116.6608	19.6361
Enthalpy of Formation	kcal/mol	0.0000	0.0000	-17.8900	19.8200	-29.4300	-57.8000
Gibbs Energy of Formation	kcal/mol	0.0000	0.0000	-12.1500	30.9900	7.5900	-54.6400
Heat of Vaporization	kcal/mol	0.2160	1.3330	1.9550	7.3520	7.1600	9.7170
Heat Capacity	cal/(mol C)						
A1		6.1038	7.4400	4.5980	-8.1010	-13.0270	8.0582
A2 x 1.0E3		3.5399	-3.2400	12.4500	113.3000	146.0000	-1.6761
A3 x 1.0E6		-5.3789	6.4000	2.8600	-72.0600	-60.2700	6.5196
A4 x 1.0E9		4.2220	-2.7900	-2.7030	17.0300	3.1560	-3.9760
A5 x 1.0E12		-1.4176	0.0000	0.0000	0.0000	0.0000	1.0265
A6 x 1.0E15		0.1704	0.0000	0.0000	0.0000	0.0000	-0.0996
Low Temperature Limit	C	-73.1500	26.8500	26.8500	26.8500	26.8500	-73.1500
Upper Temperature Limit	C	2726.8500	936.0500	1367.8500	1137.3500	1082.3500	2726.8500
Liquid Vapor Pressure	Pa						
A1		25.3527	46.1539	43.3183	73.8624	73.3379	65.1544
A2		-142.8939	-984.2006	-1380.5780	-5970.4390	-5885.9268	-6842.9072
A3		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
A4		0.0971	0.0232	0.0116	0.0055	0.0055	0.0028
A5		-2.9194	-5.4530	-4.3964	-8.0798	-8.0304	-6.1364
A6		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
A7		6.0000	6.0000	6.0000	6.0000	6.0000	6.0000
Low Temperature Limit	C	-258.6136	-209.8500	-182.4500	20.2109	19.5774	46.1166
Upper Temperature Limit	C	2726.8500	936.0500	1367.8500	1137.3500	1082.3500	2726.8500
Liquid Vapor Pressure	Pa						
A1		25.3527	46.1539	43.3183	73.8624	73.3379	65.1544
A2		-142.8939	-984.2006	-1380.5780	-5970.4390	-5885.9268	-6842.9072
A3		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
A4		0.0971	0.0232	0.0116	0.0055	0.0055	0.0028
A5		-2.9194	-5.4530	-4.3964	-8.0798	-8.0304	-6.1364
A6		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
A7		6.0000	6.0000	6.0000	6.0000	6.0000	6.0000
Low Temperature Limit	C	-258.6136	-209.8500	-182.4500	20.2109	19.5774	46.1166
Upper Temperature Limit	C	-239.9500	-146.9500	-82.5500	288.9500	280.2500	374.1500
Transport Properties							
Diffusion Coefficient	m ² /s						
Mass Transfer Coefficient	m/s						
Heat Transfer Coefficient	kcal/(m ² s C)						
Viscosity	cp						
Thermal Conductivity	kcal/(m s C)						
Solid State		0.0	0.0	0.0	0.0	0.0	0.0

Figure 1: Physical Properties of Hydrogen, Nitrene, Methane, Benzene, Cyclohexane and Water.

4.0 – Components

- H_2 – Hydrogen gas
- N_2 – Nitrogen gas
- CH_4 – Methane
- C_6H_6 – Benzene
- C_6H_{12} – Cyclohexane
- H_2O – Water

5.0 – Equipment

- Two pumps: first pump is used in the fresh benzene stream, where its pressure goes from 15 Psia to 335 Psia. Second pump is used in the liquid benzene recycle stream, where its pressure goes from 310 Psia to 335 Psia.
- Compressor: placed in the recycling gas stream, pressure goes from 310 Psia to 335 Psia.
- Mixer: Mix all the four streams into one, fresh feed hydrogen gas, fresh feed benzene and two recycling streams.
- Heater exchanger: this equipment was designed as two heaters, where the second one (hot side) receives the heat from the top one (cold side), symbolized by Stream 11. There is also a 5 Psia pressure drop inside the heater.
- Reactor: vessel where the chemical reactions took place. Chemical reactions occurred at 400°F and 315 Psia.
- Cooler: Utilizes cold water from a cooling tower nearby. It cools the stream from 157.42°F down to 120°F.
- High pressure separator followed by splitters: vessel where the first separation took place. Separation occurred at 120°F and 310 psia, with an overhead gas purge fraction of 0.1391 and a bottom liquid recycle fraction of 0.3035.
- Distillation column: where the final separation took place, and the final product was collected. Separation occurred at 19.2 bar with a product flow rate of 300 KMTA.

6.0 – Discussion and Analysis

For this problem, the flow sheet diagram was downloaded from previous homework and used in Aspen Plus. Modifications were made into the design to add a distillation column. Two design specifications were created in order to obtain the production desired of 300 KMTA of a high purity product, and the stream going into the reactor was chosen to be the tear stream.

The first specification was created to set the recycle gas stream and the purge stream ratio. As a requirement from the system, the hydrogen rich recycle gas stream had to make up the concentration of the hydrogen in order to always keep a ratio of 3.3 hydrogen to benzene concentration on the tear stream. By doing that, the purge stream was also created to discharge the excess hydrogen and other impurities in the system.

The second design specification was set as the liquid recycle been 45% of the fresh liquid benzene flow rate. A calculator was created to determine the raw material flow rate of the hydrogen and benzene stream needed in order to produce 300KMTA of cyclohexane.

The output pressure of both the pumps and the compressor were set in order to match the pressure of the fresh hydrogen feed going into the mixer. The resultant stream was sent into the heater, where was considered a pressure drop of 5 Psia since the reactor had an operation pressure set. The utility stream was added connecting both the heaters together, where the first heater was called cold side and the second one was called hot side.

The resultant stream of the heaters was then sent to the cooler where cold water was used to reduce the temperature of the stream. As the problem states, a cooling tower nearby was designed for a dry bulb temperature of 85°F with a relative humidity of 70% and an approach temperature of 10°C. A psychrometric chart [2] was used to determine the wet bulb temperature of the water as 72.2°F. It was also determined that the water could heat up to 122.2°F due to the approach temperature requirement. Three different utility streams were set up: water, electricity and steam.

For the distillation column, first the short-cut model, SEP2, was used to find the column operating pressure. Since methane was identified as our light key and benzene is the heavy key, we set the split fraction of hydrogen and nitrogen to 1 because they have smaller boiling points than the methane, and thus will only separate to the distillate. For the cyclohexane, since it has a higher boiling point than the benzene, we set that split fraction to 0.001, just like benzene, since very trace amounts will be in the distillate. We also specified an outlet flash temperature of 120°F and a vapor fraction of 1 for the fuel stream because the distillate is only vapor. Since we have hydrogen gas, it will not be feasible to liquefy the fuel stream. We are also using cooling water for the condenser, and therefore cannot transfer heat past 120°F. After running the column, we obtain a column operating pressure of 278.41 psia, or 19.2 bar.

The second step we needed to do before running the rigorous distillation was run a DISTWU column. This will give us the reflux ratio, number of stages, feed stage, and distillate to feed fraction we need for the rigorous distillation, RADFRAC. Using our previously mentioned pressure, we set that as our condenser pressure and set a reboiler pressure at 19.7 bar to account for the pressure drop in the column. Since we do not know the reflux ratio, we set the column specification to -1.15 reflux ratio, indicating that we will use $1.15R_{\min}$ as our actual reflux ratio. For the Key Component Recoveries, we set a recovery of 0.999 for methane and 0.0001 for Benzene. We also know that the distillate will be all vapor, so we set the condenser specifications to a partial condenser with all vapor distillate. The following results were calculated.

▶	Minimum reflux ratio:	0.465434	
▶	Actual reflux ratio:	0.535249	
▶	Minimum number of stages:	3.97765	
▶	Number of actual stages:	13.3406	
▶	Feed stage:	4.74602	
▶	Number of actual stages above feed	3.74602	
▶	Reboiler heating required:	1.58821e+07	Btu/hr
▶	Condenser cooling required:	136675	Btu/hr
▶	Distillate temperature:	-123.206	F
▶	Bottom temperature:	437.891	F
▶	Distillate to feed fraction:	0.0403707	

Figure 2: DISTWU Results from Aspen.

Here, we obtain 14 stages, a feed stage of 4, a reflux ratio of 0.535, and a D/F ratio of 0.04037. We input these into RADFRAC.

Once we input these components into RADFRAC, we saw that we had an error of the trays drying out. To accommodate this, we reduced the number of trays to 6. We also noticed that our fuel stream temperature was too low at 96.47°F. Since our original outlet flash temperature is 120°F, we wanted to hit a target closer to that. After adjusting the reflux ratio to 0.35, we obtained a fuel outlet temperature of 111.29°F. With the output of RADFRAC, we are now able to place pseudo-streams to calculate the cost of utilities in the condenser, reflux pump, and reboiler.

For the condenser, we know that the stream coming into it is V2, which is the vapor coming from the internal stage 2. To simulate this, we add a heater to the pseudo-stream, V2. We know that there is no pressure drop across the condenser, so

we set the pressure to 0 psia. However, we do not know the duty. Since we are simply simulating the condenser, we can place a transfer from the RADFRAC condenser duty to the duty of this simulated condenser. Next, we select the utility as water from our Utility ID option to get a cost estimate. The outlet temperature of the simulated condenser is the same as what comes out of RADFRAC, indicating that the simulation was completed successfully. We also see that the duty, -323861.024 Btu/hr, matches that from RADFRAC as well.

The reboiler was executed the same way, but instead of vapor from stage 2, we set our internal stage to stage 5. This is the liquid going into the reboiler at the bottom of the column. Again, we do a transfer block, but from the RADFRAC reboiler duty instead. Finally, the utility is set to STEAM from our Utility ID options to get the cost estimate. The outlet temperature and duty match those from RADFRAC, indicating a successful simulation.

A pseudo stream was added for the reflux being pumped back to the distillation, in order to add this stream, a reflux stream was created as an internal pseudo type at stage 1 location and internal liquid phase. To calculate the power needed for this pump, we use the formula $\Delta P = \rho * g * H$, where ρ is the density of the fluid (46.1442 lb/ft³) pumped back to the distillation column, g is the gravity constant (9.81 m/s²) and H is the height of the column that the fluid must overcome, (10.786m). The pump was set up to have the pressure increased of 0.79 bar.

Water utility was set as $\$1.198 \times 10^{-5}$ per lb and a cooling value of -30 Btu/lb. Electricity was set as \$0.07 per kW/h. Finally, steam was set as \$6.48 per Mlb and 1713.5 kJ/Kg. Utility cost was calculated using Aspen and the results are displayed in Appendix III.

To finalize this problem, a personalized set of units was created in Aspen, so the chart displaying the results would have the KMTA units for the flow rate.

7.0 – Appendix

7.1 – Appendix I: Formulas and Chart

Since this problem is the only problem involving chemical reactions, the following were used for mass balance of the compounds

$$\text{Hydrogen: } n_{H_2} = n_{H_2}^0 - 3 * \mathfrak{Z}_1 - 9 * \mathfrak{Z}_2$$

$$\text{Nitrogen: } n_{N_2} = n_{N_2}^0$$

$$\text{Methane: } n_{CH_4} = n_{CH_4}^0 + 6 * \mathfrak{Z}_2$$

$$\text{Benzene: } n_{C_6H_6} = n_{C_6H_6}^0 - 1 * \mathfrak{Z}_1 - 1 * \mathfrak{Z}_2$$

$$\text{Cyclohexane: } n_{C_6H_{12}} = n_{C_6H_{12}}^0 + 1 * \mathfrak{Z}_1$$

- Conversion of Benzene

$$\mathfrak{Z}_1 + \mathfrak{Z}_2 = n_{C_6H_6}^0 * \text{conversion}$$

Where $\mathfrak{Z}_1$ and $\mathfrak{Z}_2$ are coefficients of reaction 1 and reaction 2 respectively.

Cost Analysis:

- Compressor

Equipment cost

$$\$ (2010) = 580,000 + 20,000 * (W(kW))^{0.6}$$

Installation Cost

$$\$ (2010) = 2.5 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Where,

$W(kW)$ is the power of the compressor in kW.

- Heaters / Cooler

$$\Delta T_{\ln} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$A = \frac{Q}{F_t * U * \Delta T_{\ln}}$$

Equipment cost

$$\$ (2010) = 28,000 + 54 * (A(m^2))^{1.2}$$

Installation Cost

$$\$ (2010) = 2.0 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Where,

A is the heat transfer area in m².

Q is the duty of the heater.

F_t is 0.8 factor.

U is the heat transfer coefficient, which is $50 \frac{Btu}{h * ft^2 * F}$ in this case.

- Pump

Equipment cost

$$\$ (2010) = 8,000 + 240 * (Q(\frac{l}{s}))^{0.9}$$

Installation Cost

$$\$ (2010) = 4.0 * \text{Equip. Cost in 2020}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Where,

$Q(\frac{l}{s})$ is the flow rate in liters per second.

- Vertical Vessels

Assuming 5 minutes of residence time

Equipment cost

$$\$ (2010) = 11,600 + 34 * (M(Kg))^{0.85}$$

Installation Cost

$$\$ (2010) = 4.0 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Operating Pressure

$$\text{Thickness} = \frac{P_i * D_i}{2SE - 1.2 * P_i}$$

$$\text{Mass} = \pi * D_i * L * \text{thickness} * \rho$$

Volume of the vessel

$$V_{\text{vessel}} = \frac{\pi * D_i^3}{2}$$

Where,

M(Kg) is the total mass of the vessel in Kg.

P_i is the internal operating pressure in Psig.

D_i is the internal diameter.

E is the welded efficiency = 1

S is the stress of the vessel = $88.9455 \frac{N}{mm^2}$

L is the length of the vessel.

P is the density of the material, stainless steel is $8000 \frac{Kg}{m^3}$

- Distillation Column

- Trays

$$a = \frac{K(lk)}{K(hk)}$$

$$a = \sqrt{a(\text{top}) * a(\text{bottom})}$$

$$\text{Efficiency} = 50.3 * (a * \mu)^{-0.226}$$

$$\text{Num of Trays} = \frac{\text{Number of stages}}{\text{Efficiency}}$$

$$H = 1.15 (\text{number of trays} * 0.6m)$$

Equipment cost

$$\$ (2010) = 130 + 440 * (\text{Diameter}(m))^{1.8}$$

Where,

K_{lk} and k_{hk} are the k values of the light key and the heavy key components respectively.

μ is the viscosity of the mixture

H is the height of the column in meters

- Vessel

Operating Pressure

$$\text{Thickness} = \frac{Pi * Di}{2SE - 1.2 * Pi}$$

$$\text{Mass} = \pi * Di * L * \text{thickness} * \rho$$

Assuming 5 minutes of residence time

Equipment cost

$$\$ (2010) = 11,600 + 34 * (M(\text{Kg}))^{0.85}$$

Installation Cost

$$\$ (2010) = 4.0 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

- Reboiler

$$\Delta T_{ln} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$A = \frac{Q}{F_t * U * \Delta T_{ln}}$$

Equipment cost

$$\$ (2010) = 29,000 + 400 * (A(\text{m}^2))^{0.9}$$

Installation Cost

$$\$ (2010) = 4.0 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Where,

A is the heat transfer area in m^2 .

Q is the duty of the reboiler.

F_t is 0.8 factor.

U is the heat transfer coefficient, which is $150 \frac{\text{Btu}}{\text{h} * \text{ft}^2 * F}$ in this case.

- Condenser

$$\Delta T_{\ln} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)}$$

$$A = \frac{Q}{F_t * U * \Delta T_{\ln}}$$

Equipment cost

$$\$ (2010) = 28,000 + 54 * (A(m^2))^{1.2}$$

Installation Cost

$$\$ (2010) = 4.0 * \text{Equip. Cost in 2010}$$

Annual Cost

$$\$ (2010) = \frac{4}{3} * \text{installation cost in 2010}$$

Where,

A is the heat transfer area in m².

Q is the duty of the heater.

F_t is 0.8 factor.

U is the heat transfer coefficient, which is $100 \frac{Btu}{h * ft^2 * F}$ in this case.

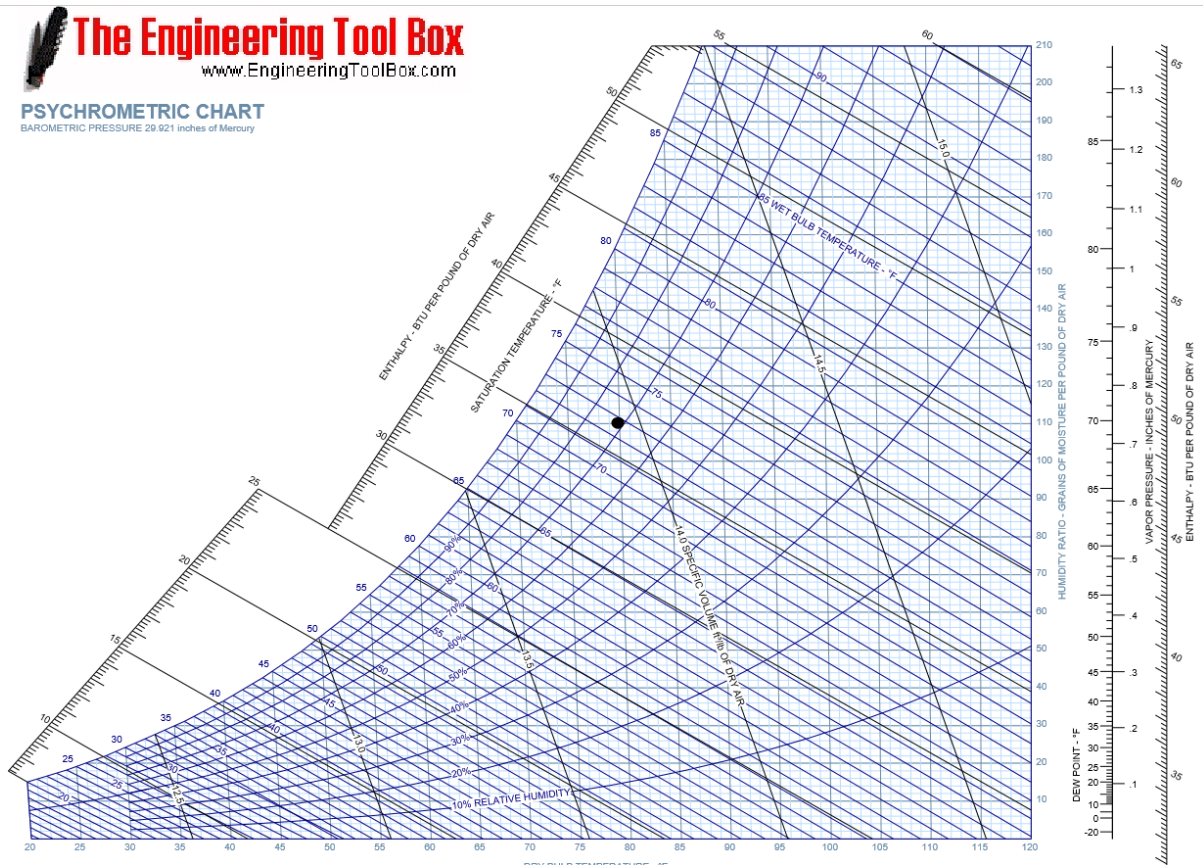


Figure 3 [1]: Psychrometric Chart for finding the wet bulb temperature of water

7.2 – Appendix II: Flow Diagram

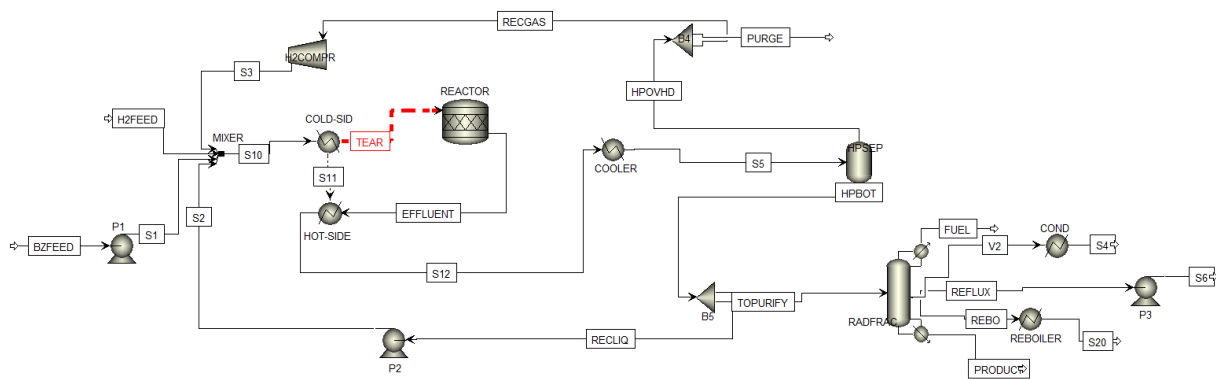


Figure 4: Flow Diagram for Cyclohexane Production, created on Aspen Plus.

7.3 – Appendix III: Results

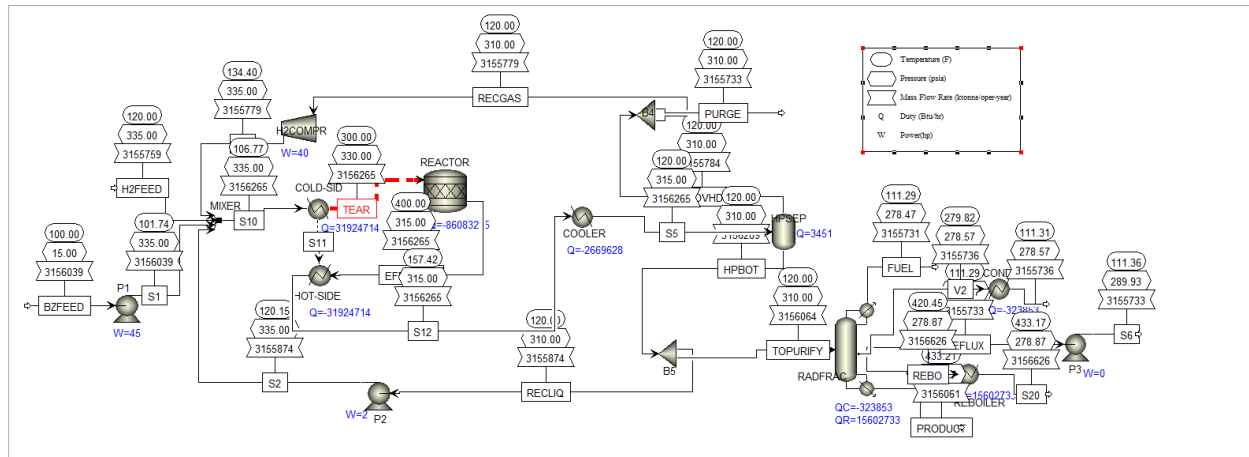


Figure 5: Complete flow diagram with results

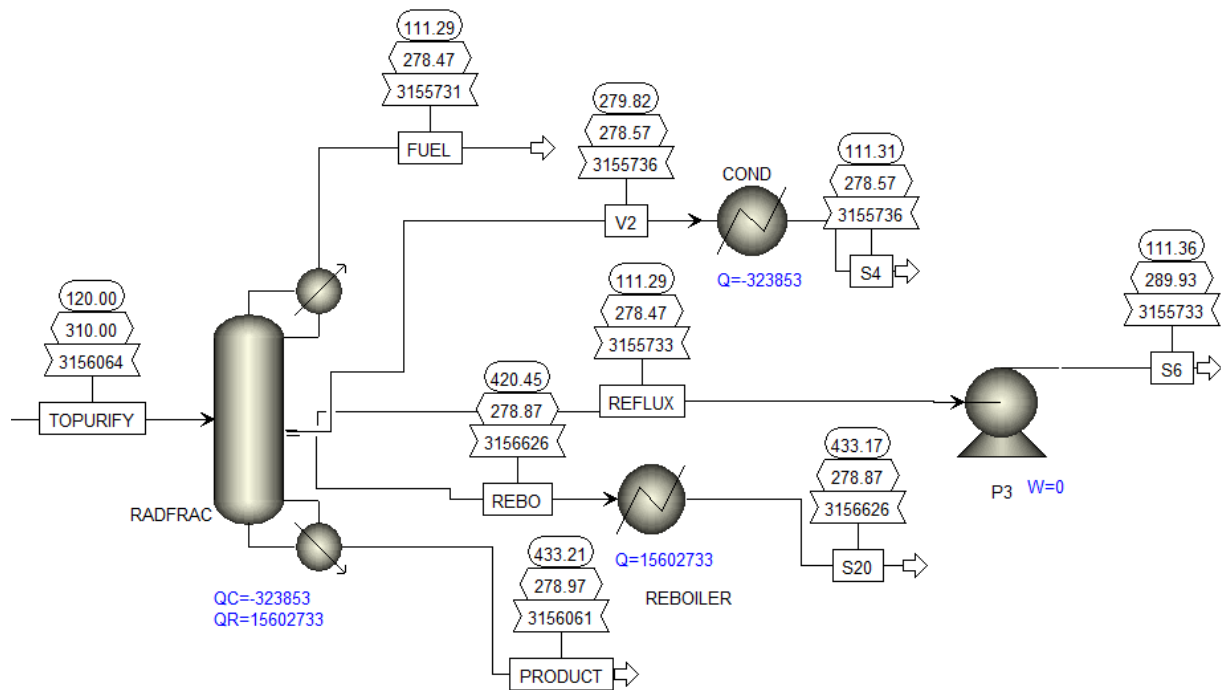


Figure 6: Results with focus on the Radfrac and pseudo streams.

Cyclohexane Production Cost Estimate						
EQUIPMENT LIST: CAPITAL COSTS						
	Project: Assignment #5		Chemical Engineers: Mariana St. Peter	Jasmine Villegas		
	Location: Chemical Engineering Department		Date: 11/4/2019			
Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
Drums						
Vertical Tank	HPSEP	High Pressure Separation	V-1	\$153,697	\$614,790	\$819,720
Heat Exchangers						
S&T Exchanger	HOT-SIDE	Heats Reacting Components	E-1	\$248,676	\$870,366	\$1,160,488
S&T Exchanger	COOLER	Cools Reactor Effluent	E-2	\$248,676	\$870,366	\$1,160,488
Water Coolers						
Water Cooler	--	Cools Reactor Effluent	EA-2	\$195,123	\$682,931	\$910,575
Pumps						
Pump	FBPUMP	Fresh Benzene Pump	P-1	\$10,155	\$40,621	\$54,161
Pump	RBPUMP	Recycle Benzene Pump	P-2	\$9,247	\$36,988	\$49,318
Pump	RFPUMP	Reflux Distillation Pump	P-3	\$8,057	\$32,226	\$42,968
Compressors						
Compressor	H2COMPR	Compress the Recycling Gas	C-1	\$733,393	\$1,833,482	\$2,444,642.08
Distillation Column						
Trays	--			\$1,433	\$5,733	\$7,643.71
Column	RADFRAC	Column used to separate the product from the fuel gas		\$103,680	\$414,719	\$552,958.72
Reboiler	REBOILER	Reboiler used to boil the liquid phase inside the column		\$129,148	\$516,593	\$688,791.02
Condenser	COND	Condenser used to condense the vapor phase of the distill		\$34,693	\$138,771	\$185,028.53
Total				\$1,875,978	\$6,057,587	\$8,076,782

Figure 7: Result for the capital costs of equipment for the cyclohexane production.

Main Flowsheet x WATER (UTILITY) - Results x +						
Results State Variables Status						
Block ID	Block type	Duty Btu/hr	Usage ktonne/op	Cost \$/hr	CO2 emission rate	
COND	HEATER	323853	3.15577e+06	0.129325		
COOLER	HEATER	2.66963e+06	3.15608e+06	1.06607		

Figure 8: Results for water usage

Main Flowsheet x ELECTRIC (UTILITY) - Results x +						
Results State Variables Status						
Block ID	Block type	Duty Btu/hr	Usage kW	Cost \$/hr	CO2 emission rate	
P1	PUMP	114867	33.6641	2.35649		
P2	PUMP	4891.94	1.43369	0.100358		
P3	PUMP	72.1596	0.0211479	0.00148035		
H2COMPR	COMPR	102307	29.9832	2.09882		

Figure 9: Results of electricity usage

Main Flowsheet STEAM (UTILITY) - Results						
Results State Variables Status						
	Block ID	Block type	Duty Btu/hr	Usage ktonne/op	Cost \$/hr	CO2 emission rate
	REBOILER	HEATER	1.56027e+07	3.15581e+06	137.247	

Figure 10: Result of steam usage

Since at the reboiler, the stream is at 278.97 psia, which is equivalent to 19.22 barg, is recommended to use high pressure steam [5] since this value is above 17.5 barg.

Cyclohexane Production Cost Estimate						
UTILITIES: MATERIAL COST						
	Project:	Assignment #5	Chemical Engineers:	Mariana St. Peter	Jasmine Villegas	
	Location:	Chemical Engineering Department	Date:	11/2/2019		
Utility Class	ASPEN ID	Service	Tag	Usage per hour [lbmol/h]	Unit Cost	Annual Cost, \$/yr
Hydrogen	H2FEED	Hydrogen Feed	HF-1	3038.96	\$-2.60	-\$23,979.517
Benzene	BZFEED	Benzene Feed	BF-1	999	-\$760.00	-\$215,199.849
Methane	FUEL	Fuel gas by-product	FG-1	1152400	\$7.30	\$67,300
Cyclohexane	PRODUCT	Cyclohexane product	CH-1	992.133	\$1,040.00	\$315,112.191
					Total Utility Cost:	\$291,132.674

Cyclohexane Production Cost Estimate						
UTILITIES: OPERATING COST						
	Project:	Assignment #5	Chemical Engineers:	Mariana St. Peter	Jasmine Villegas	
	Location:	Chemical Engineering Department	Date:	11/2/2019		
Utility Class	ASPEN ID	Service	Tag	Usage per hour	Unit Cost	Annual Cost, \$/yr
Cooling Water [kJ]	--	Water used in the cooler	--	208174	\$0.0800	\$9,563
Steam [kJ]	--	Steam used to heat up the reboiler	--	793677387.7	\$11.4500	\$1,097,976
Power [kw]	--	Power supplied to pumps and compressors	--	65.103	\$0.0900	\$36.457
					Total Utility Cost:	\$1,143,996

Figure 11: Summary of the operating cost results

7.0 – References

- [1] Engineering ToolBox, (2004). *Psychrometric Chart - Barometric Pressure 29.921 inches of Mercury*. [online] Available at: https://www.engineeringtoolbox.com/psychrometric-chart-d_816.html [Accessed 21 Oct. 2019].
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- [4] Towler, G., & Sinnott, R. (2008). *Chemical Engineering Design* (2nd ed.). Burlington, MA: Elsevier.
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